Growing up across 5 countries with various languages, I've long been fascinated with language. Over time, as a high-schooler developing a hobby for reading papers on arXiv, this fascination (and finding RNNs very cool) drew me towards NLP, first as a reader, and eventually getting to do research myself. Now, I've become interested in everything related to better understanding and applying language models: from efficient decoding, to analyzing post-training, to studying sociolinguistic variation. To narrow things, I'll discuss two questions I've been thinking about recently:

(i) What do generation models actually learn during training?

(ii) Which properties of data play a role in building better systems?

Measuring what models actually learn, as they learn

Spurious correlations have long plagued generalization across many tasks (e.g. NLI) in NLP, with "improvements" often relying on shallow data artifacts, and since my first project, studying ways to interpretably predict model generalization **(Singhal et al. 2023)**, I've been interested in these phenomena. NLP now focuses on text *generation*, but despite great reported progress, there's limited understanding of what underlies this progress; it's unclear whether spurious correlations still play a role today. Last year, I started studying this in reinforcement learning from human feedback (RLHF), which involves (1) learning a reward score that correlates with human preferences and (2) maximizing reward of a generation model.

My experiments found reward models correlated strongly with length, and RLHF noticeably increased model output length, suggesting a worrying hypothesis: RLHF may *only* optimize for length, and nothing else. Ultimately we found the hypothesis to be true (**Singhal et al. 2024, COLM Oral**): length accounted for over 90% of optimization on several settings, and *length-only* optimization reproduced the downstream gains of full RLHF. This behavior, which we traced to minor data biases, remained even across a variety of anti-length interventions.

Humans like longer outputs, so this isn't inherently bad, but our work revealed a larger issue: generation *is* prone to shallow optimization, and human-centric evaluation *isn't* equipped to reveal when this happens, especially on unexpected features. Beyond benchmarks, or post-hoc interpretation, my hypothesis is thus that, in order to understand and address this, we need tools that can discover and accurately measure what behavior models actually learn from training.

From tokenwise losses, to parameter changes, to checkpoint output distributions, we have access to a great deal of *training-centric information* that could enable such tools. For example, in the length study, we found comparing RLHF reward changes across checkpoint outputs while controlling for length allowed us to approximate how much length contributed to learning between checkpoints. However, length is only one feature, and many others may exist. I hypothesize that, given training-centric information, we can "discover" these, maybe by (A) filtering down from a large generated feature space (e.g. bag-of-words, graph representations, natural language descriptions), or (B) training systems that can automatically infer what models learn. This could be evaluated by "planting" simple features in data and measuring whether we can re-discover them, or seeing whether feature-only optimization reproduces standard training.

This direction excites me because it enables precise measurement of what models learn without prior assumptions, enabling a re-analysis of RLHF, fine-tuning, or even pre-training by simply measuring some training jobs. Such tools could measure evaluation game-ability (e.g. how much improvement is explained by simple features), or help balance/control test sets against simple features. Likewise, this sort of breakdown would be invaluable for practitioners both as a debugging tool, or enabling clearer iteration towards mitigating optimization of undesired features to improve model robustness.

Principles For Improving Data and Data Structure:

Data is a crucial ingredient in NLP, yet even as a research intern at Scale AI, I've found data to often show problems upon closer scrutiny. For example, after ~5 minutes of looking through a "high quality" dataset from LIMA (Zhou et al. 2023), I found data describing Roman architecture when asked about the Roman government. Given the successes we've seen recently, I wonder how such problems in data affect models, and if models are robust to them, what then actually matters in data?

After our RLHF/length work, we started examining conditions for robust preference learning, finding, even with data interventions, that on similar "gold" preference data (e.g. which outputs to calculator-style questions are closer to correct), DPO, RLHF and other algorithms usually perform similarly. However, as we explored collecting preferences *online* (e.g. instead of collecting 10k gold preferences at once, iteratively collect/train on 10 preferences at a time, 1k times), we started finding large improvements on many tasks **(Singhal et al. 2024, COLM)**. This confirmed to me that, beyond scaling, differences in data *can* greatly affect what models learn.

Many scientific questions here excite me: (i) Given a task, can we actually measure "data quality", and does it relate to intuition? For example, can we develop heuristics or trained metrics to predict how well models will learn tasks from batches of data? (ii) What determines whether models learn to hallucinate or generalize from a datapoint? This may relate to shallow priors, parametric knowledge, or something else, but we could potentially test by constructing data with different properties. (iii) How does inference relate to the structure of data? Since working on efficiently processing large output sets with transformers **(Singhal et al. 2023),** I've wondered how analyzing outputs could better understand models. For example, does the output space given a prompt (e.g. 5k sampled outputs) relate to co-occurrence or frequency patterns in training data, and can analyzing training data inform better decoding strategies? Likewise, can we augment existing data, or even generate new data that can precisely edit model behavior with minimal supervision?

Career Goals: I want a PhD because it will give me an extended period to learn and focus on improving as a scientist. My long-term goal is to become a professor. My advisor Greg Durrett is a major inspiration to me, and if possible I really want to reproduce some of the immense positive impact he's had on me. This follows other reasons: (a) I really enjoy teaching / mentorship, (b) I want to help the NLP community become a better place to do *open* science, (c) I realized from my time at Scale that industry's time, direction, and approach pressures can align less with the understanding-driven style of research I want to do.

UC Berkeley There are several faculty at UC Berkeley I'd be excited to work with, specifically Professors Jacob Steinhardt, Sewon Min, Alane Suhr, and Dan Klein. I especially appreciate the diverse tasks and directions pursued within NLP groups at Berkeley. Following work from these groups closely aligning with my interests, I think Berkeley would be a great place to do my PhD.

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